

HMGCC Co-Creation Proposal

Smart Personal Assistant for Security Researchers

Submitted by OTShield | Research-First. Offline-Ready. Evidence-Driven.

Organisation	Safetech Global Limited (trading as OTShield)	Reg. No.	15233187
Primary contact	Fatma Erturk, Project Lead fatma.erturk@otshield.io 07453 641903	Address	7 Bell Yard, London, WC2A 2JR
Web / LinkedIn	https://otshield.io/ @ot-shield @the-ot-hacker-mindset	Funding	£60,000 all-inclusive 12 weeks TRL 6
Demo Video	https://www.youtube.com/watch?v=2QdR3wRGQhk		

1. Scope & alignment with the challenge

Security researchers spend more time **finding, indexing and interpreting** technical data than acting on it; interpretation of complex artefacts is the most time-intensive stage in HMGCC's recent projects. Across any complex hardware / software domain HMGCC's tear-down workflows touch (ICS is the worked example, not the limit), this bottleneck delays vulnerability discovery, especially when the target is heterogeneous and operated in air-gapped facilities.

Our proposal: extend the existing OTShield platform, an offline, evidence-driven, copilot-ready security platform with deep ICS pedigree, into a TRL 6 research-assistant capability applicable to any complex hardware / software tear-down, that meets every essential HMGCC requirement and addresses all four desirable requirements.

Why us: we are not building a better search tool. We are building a *cognitive layer for vulnerability research* that does what RAG cannot, bridging documentation with real network behaviour through OTShield's live PCAP, decoys and asset inventory. Research Mode is operational at TRL 3 to 4 today; the supporting SOC-side decoy fabric (TRL 5, validated against real attackers) feeds it via signed update bundles. The 12-week project takes Research Mode to TRL 6 with HMGCC test data; today-vs-Phase-1 delta in Section 2.

Dual-use by design, HMGCC will not be a sole customer. OTShield is a commercial ICS security product targeting critical infrastructure operators, defence supply chain and ICS vendor red teams; pre-revenue with an active pipeline and three published price tiers. Safetech Global (inc. Oct 2023) and OTShield hold **12 industry awards** and have been confirmed by UK Enterprise Awards as **Most Innovative OT Security Company 2026** (under embargo, July 2026; see Annex C). HMGCC inherits a commercially supported tool with a Phase 1 royalty-free licence.

OTShield today: live capabilities (the dual-use evidence)

Live capability	What it does today, and why it matters for the challenge
Research Mode	Bundle-based research workspaces (Library, Summary, Inventory, Ports & Services, Threads, Findings, Vulns) on a Java + Spring Boot backend with offline Ollama LLM (Llama 3.2 1B today) and an in-memory vector store. Foundation for the HMGCC challenge.
SOC Mode + Decoy layer	Realistic protocol-aware decoys for Modbus, S7Comm, EtherNet/IP, OPC UA, DNP3 and more, deployed today on an internet-exposed decoy fabric operated by Safetech Global and capturing real attacker behaviour against ICS protocols.
PCAP, asset inventory + MITRE ATT&CK for ICS	Purdue-aware packet processing with dedicated dissectors for SCADA communication protocols; auto-discovered PLCs / RTUs / HMIs with risk scoring; attacker TTPs mapped across the full ATT&CK for ICS kill chain; IOC export with SIEM / TAXII / MISP push. Lets the assistant validate documentation against on-wire behaviour.
Live ICS threat-intelligence platform	Internet-exposed decoy fabric operated by Safetech Global, capturing real ICS attacks: in the first days of live operation alone, 157 unique attackers from 20 countries executing 12,968 attacks (script-kiddie scans to advanced actors targeting ICS protocol weaknesses). Packaged as signed update bundles for the air-gapped tool. <i>Full breakdown in Annex B.</i>
Commercial maturity & regulatory alignment	Three published price tiers + active customer pipeline; aligned with NIS2, IEC 62443, UK NCSC CAF ; coexists with Dragos / Clarity; deployable on Cisco hardware or software. <i>Full partner-channel evidence in Annex E.</i>

Mapping to the three challenge objectives: (i) **Index structured + unstructured data:** PDF + plain-text today via Apache PDFBox; multi-modal ingest (OCR, handwriting, schematic) added in Sprint 1. (ii) **Clear technical summaries:** local Ollama LLM (Llama 3.2 1B today, Gemma 4 in Phase 1) produces per-bundle Summary / Inventory / Ports & Services views. (iii) **Source-cited Q&A:** Threads tab with citations + HIGH/MED/LOW + "needs more sources" flag (Phase 1: calibrated conformal score), promote-to-Findings, adaptive profile.

2. System approach & innovation

The tool is a locally-hosted web app on the researcher's laptop (browser UI from localhost; backend, local LLM, embeddings, indices on the same machine; no internet calls). The backend is **Java 21 + Spring Boot** wrapping a five-stage pipeline: **(1) Ingest** → **(2) Index & extract** → **(3) Retrieve** → **(4) Reason & verify** → **(5) Answer & ledger**.

Today's stack vs. Phase 1 upgrade (Research Mode: TRL 3 to 4 → TRL 6)

Layer	Today (operational, in production code)	Phase 1 upgrade
LLM serving (3-tier)	Ollama at localhost:11434, Llama 3.2 1B Instruct Q4_K_M (2K context, top-K=4, min relevance 0.18)	Tier 1 / Edge laptop (4 to 8 GB VRAM, primary deployment): Gemma 4 9B Q4_K_M for 4 GB-class laptops, Gemma 4 13B Q4_K_M for 8 GB-class laptops; Llama 3.2 1B retained as CPU-only / ablation fallback. Tier 2 / HMGCC OpenAI-style endpoint (optional scale-up): any open-weights model of HMGCC's choice (e.g. Llama 3.3 70B, Qwen 2.5 72B, Gemma 4 27B). Used by the L2 verifier for higher-accuracy cross-checking and by an automatic escalation path when edge confidence falls below threshold. Air-gap preserved (HMGCC-hosted infrastructure). Tier 3 / OTShield dev workstation (RTX 5090 32 GB, internal only): Gemma 4 26B MoE / 31B Dense Q5_K_M for benchmark-driven model selection and conformal calibration; not part of the deliverable.
Embeddings	nomic-embed-text via Ollama (768-dim, English-centric)	BGE-M3 added (BAAI, MIT, 100+ languages, 8K input, dense + sparse + multi-vector); nomic kept for English-only fast path
Vector store	InMemoryVectorStore (ConcurrentHashMap, brute-force cosine, < 500 chunks, < 1 ms search)	ChromaDB (Apache 2.0, embedded, persistent, hybrid sparse + dense); existing in-memory store kept as a fast in-process cache
Retrieval	Single-modal (vector cosine only)	Hybrid retrieval: dense (BGE-M3 + nomic) + sparse (BM25 over Chroma) + ICS-protocol-aware filters over structured Inventory/Ports/Services tables; cross-lingual expansion at query time
Verifier (L2)	None. Answers returned with retrieved-chunk citations only.	New IP. Independent agentic verifier on LangGraph (MIT). Three parallel checks per claim: (i) corpus re-query, (ii) trusted-source cross-check, (iii) on-wire cross-check against OTShield PCAP / decoy data (unique to us). Adversarial validation by Jon Medvenics with PyRIT . Per-claim decision log labels every output as known , inferred or uncertain for human-readable audit.
Confidence (L3)	HIGH / MEDIUM / LOW label + binary 'needs more sources' flag emitted by the assistant on every answer; cosine relevance per retrieved chunk (not user-facing as a calibrated 0 to 1 confidence yet)	New IP. Calibrated 0 to 1 per claim using conformal prediction (primary) + Platt scaling / isotonic regression (baselines). 'Needs more sources' flag below threshold (default 0.6, HMGCC-tuneable).
Multi-modal ingest (L1)	PDF text + plain-text fallback (markdown, text, CSV).	Full HMGCC format coverage: Word (.docx), Excel (.xlsx), PowerPoint (.pptx) via Apache POI; JPEG / PNG / BMP / TIFF, schematics, draw.io outputs (.drawio XML), hand-drawn and scanned material via Tesseract 5 + PaddleOCR + TrOCR (handwriting) + LayoutLMv3 + Gemma 4 vision; protocol-aware extractors for Modbus, S7Comm, EtherNet/IP, OPC UA, DNP3, IEC 60870-5-104.
Multilingual	Offline output translation in production for EN, TR, DE, FR, ES via the local Ollama chat model with a domain-tuned translation prompt that preserves citations, CVE ids, port numbers, IPs and ICS acronyms; LRU-cached per session.	NLLB-200 distilled-600M upgrades the chat-model translator and prioritises HMGCC's stated languages (DE, JA, FR, ZH); bilingual technical-term dictionary curated with HMGCC; cross-lingual retrieval (EN query, non-EN source) over BGE-M3.
Adaptive profile (L4)	Threads + Findings persistence.	New IP. Cross-bundle long-term memory schema; learns researcher terminology and preferred output format; proactive surfacing of related findings.

Innovation, beyond off-the-shelf RAG. Some applicants will ship a conventional document-Q&A system. We address all four limitations HMGCC raised in Q34 / Q35: (i) **structured representation alongside Q&A**, the LLM extractor produces a **component knowledge graph** (SBOM-style hierarchies, firmware memory maps, communication-interface graphs, attack-surface relationships) in parallel with the vector embeddings, so the assistant supports both "show me what you found" (graph view) and "what does this device do" (Q&A view); (ii) **on-wire cross-check**, verifier validates documented protocol behaviour against real PCAP traffic and decoy interactions from OTShield's platform (no other applicant has this); (iii) **multilingual + cultural-bias awareness** with four-layer bias mitigation, ingestion-time tagging and cross-lingual retrieval (see desirable c); (iv) **robust offline updates**, twofold signed-bundle pipeline with cryptographic integrity verification (see desirable d). Calibrated confidence using **conformal prediction** sits across all four.

IP position. *Existing (no charge to HMGCC):* Java + Spring Boot Research / SOC platform, Ollama integration, in-memory vector store, deep-extract pipeline, protocol-aware extractors, PCAP / decoy / asset platform, live threat-intel feed. *New IP generated in Phase 1:* agentic verifier with on-wire cross-check, calibrated ICS confidence model, multilingual / cross-lingual layer, long-term memory schema, hybrid retrieval over ChromaDB. All new IP owned by OTShield; Authority granted a non-exclusive royalty-free licence per HMGCC Co-Creation Terms and Conditions (T&Cs). *Full module-by-module specs in Annex A.*

3. Deliverables

The 12-week MVP is a locally-hosted web application running entirely on the researcher's laptop, browser-accessed at localhost, no internet dependency. Shipped with documentation, a reproducible test corpus and a verification report.

HMGCC essential requirement (8/8)	OTShield delivery in Phase 1	Status
Understand system architecture (interfaces, data, protocols)	Inventory + Ports & Services + Network Topology; LLM-driven inventory deep-extractor today, plus dedicated dissectors for SCADA communication protocols. Phase 1 extends with additional protocol-specific document extractors. Architectural-insight extraction : high-level component design (firmware modules, communication interfaces, trusted / untrusted domains, external dependencies and attack-surface enumeration) generated by the LLM extractor and persisted as a queryable component knowledge graph for both Q&A and graph-view access.	Met
Validate responses, prevent hallucinations	L2 verifier agent with user-selectable mode : (a) automated self-check with mandatory pre-publication validation gate, or (b) source-listing mode where the assistant lists cited sources for human review without auto-blocking. On-wire cross-check + decision log apply in both modes. Failure-mode mitigations : assistant explicitly recognises "Not enough information" rather than fabricating; prompts the user to continue, expand sources or refine; supports interrupting in-flight reasoning when the prompt was lacking.	Met
Characterise multimedia inputs (manuals, schematics, datasheets, images, code, handwriting)	Today: PDF, markdown, text, CSV. Phase 1 Sprint 1 covers the full HMGCC test-data list: Word (.docx), Excel (.xlsx), PowerPoint (.pptx), JPEG / PNG / BMP / TIFF, schematics, draw.io (.drawio XML), hand-drawn and scanned material, plus Markdown / Obsidian and OneNote (.one) note-taking imports; structured database exports (CSV / SQL dumps / XML) via Apache POI + Tesseract 5 + PaddleOCR + TrOCR + LayoutLMv3 + Gemma 4 vision. Baseline : high-confidence text and label extraction. Bonus : spatial awareness on schematic diagrams (component connectivity, handwritten annotation linked to its target element); reported with explicit confidence levels rather than asserted as full visual grounding.	Met
Verify by listing sources, cross-check against high-confidence data	Citations on every answer; trusted-source registry; evidence ledger with audit trail; per-claim decision log labels each output as known , inferred or uncertain for human-readable audit.	Met
Confidence score + flag when more sources needed	Today: HIGH / MEDIUM / LOW confidence label and a binary 'needs more sources' flag emitted on every answer. Phase 1 upgrades to a calibrated 0 to 1 score per claim using conformal prediction. When the system cannot confidently resolve a query, three behaviours combine : (a) explicitly defer ("not enough information to answer with confidence"); (b) suggest further avenues of investigation (Cultural-Gap checklist, e.g. "add Dutch advisories"); (c) highlight gaps in available data (coverage label, e.g. "Coverage: English 90% / Dutch 0%").	Met
Operate on a laptop with no internet	Fully air-gapped; local Ollama LLM, local embeddings, local watch-folder ingest. 100 % offline by design.	Met
Easy, intelligent chat-style query	Threads tab: copilot with citations, follow-up, and multiple plausible hypotheses with supporting evidence when ambiguity or contradiction is detected, rather than a single 'collapsed' answer. Default: surface the top hypothesis with cited evidence; if confidence is split or contradictory sources exist, present alternatives side-by-side with their respective evidence.	Met
Persistent memory across weeks of conversation	Bundles + Threads + Findings persist locally; long-term memory schema added in Sprint 2.	Met

Desirable requirements (4/4 addressed): (a) **Adaptive user profile** delivered in Phase 1 (L4, Sprint 2). (b) **Translate / index non-English data**: offline output translation in production for 5 languages (EN, TR, DE, FR, ES) via local Ollama chat-model with domain-tuned prompt. Phase 1 extends this with NLLB-200 NMT packaged for air-gapped install, prioritising **German, Japanese, French and Chinese** per HMGCC guidance; bilingual technical-term dictionary curated with HMGCC + cross-lingual retrieval (EN query, non-EN source). (c) **Cultural-bias mitigation (four-layer + policy modes)**: (1) ingestion-time lang/origin tagging with modal-banner alert; (2) corpus-level audit with follow-up checklist; (3) retrieval-time coverage label (e.g. 'English 90% / Dutch 0%') + Cultural-Gap tab; (4) proactive UI prompt invoking the offline MT engine (already in production for EN/TR/DE/FR/ES, NLLB-200 in Phase 1) to surface alternative-cultural sources inline. User-selectable **PROACTIVE / SUGGESTIVE / OFF** policy. Aligned with NCSC RTAU bias-mitigation guidance. Full UI/UX flow in Annex A.12. (d) **Offline update mechanism (twofold)**: (i) **software / algorithm updates** (runtime, UI, model weights) and (ii) **data repository updates** (corpus, CVE feeds, vendor bulletins, threat-intel signatures from OTShield's live decoy fabric) both delivered as **signed tar.zst bundles** with **cosign** signatures, transferred via removable media or watch-folder; integrity verified before install; the laptop never connects to the internet but stays current.

Out of Phase 1 scope: binary firmware reverse engineering and source-code static analysis, per HMGCC Q14 / Q21. The assistant works from documentation about hardware / software systems (datasheets, schematics, manuals, vendor bulletins), not from compiled binaries or source. Hardware-in-the-loop firmware extraction (JTAG / SWD) is a Phase 2 candidate (see Sec. 8).

Concrete week-12 deliverables & success metrics

- Locally-hosted web app packaged for offline install on **Windows 11 and Ubuntu (latest)** laptops; signed binaries; browser UI from localhost; zero internet calls. **Two delivery modes**: (a) **physical laptop pre-installed** by Safetech Global at handover, or (b) **vSphere-compatible image** uploaded to HMGCC's existing vSphere infrastructure for both initial install and subsequent signed update bundles. Modular components are integratable into HMGCC's existing air-gapped agent platforms.
- Reference test corpus (HMGCC-supplied) + reproducible benchmark report against these **quantitative success criteria** (full methodology in Annex A.8):

- **Hallucination rate $\leq 5\%$** (verifier-flagged / contradicted claims, 200-claim adjudicated sample); **Confidence calibration ECE ≤ 0.10** (10-bin, conformal); **Retrieval precision@5 ≥ 0.80** on ICS-protocol queries.
- Stretch (tracked through Phase 2): **Cross-lingual recall@10 ≥ 0.65** (HMGCC priority languages); **latency ≤ 30 s** Tier 1 edge laptop (4 to 8 GB VRAM) / **≤ 8 s** Tier 2 HMGCC endpoint (verifier escalation); **OCR + handwriting accuracy ≥ 0.92 ; "needs more sources" precision ≥ 0.85 .**
- **Researcher saved-time assessment (qualitative)**: at the three sprint-review demos (Wk 4, 8, 12), HMGCC security researchers compare time-to-first-finding using the assistant against their current tooling on a representative tear-down task; net positive saved-time on at least 2 of 3 sessions is a Phase 1 outcome target.

The first three quantitative criteria are the minimum acceptance criteria for week-12 sign-off; the rest are stretch goals. The qualitative saved-time assessment aligns with HMGCC's stated evaluation method.

- Technical documentation (requirement-discovery log, architectural design decisions, threat model, data-handling plan, test methodology and results, operator guide, IP register, signed CycloneDX SBOM with pinned versions); code handover (Git repository archive, README, build instructions, dependency manifest, sample queries) to a standard suitable for review and ongoing maintenance by HMGCC engineers; TRL 6 evidence pack (benchmark harness, raw measurements, calibration plots, adjudication log, signed-off methodology); Phase 2 roadmap with costings (Sec. 8).

4. Co-creation model: how we will work with HMGCC

We treat HMGCC as an embedded partner shaping the product across all 12 weeks. The engagement model below is open to refinement at kick-off.

Engagement	Cadence	Purpose & co-created output
Kick-off on-site at HMGCC Milton Keynes	Wk 1, 1 to 2 days	Architecture & threat-model walkthrough; align on trusted-source taxonomy, confidence thresholds, acceptance criteria. Output: signed-off Phase 1 plan.
Weekly delivery sync	30 min, weekly	HMGCC delivery manager + OTShield tech lead. Risks, blockers, scope trade-offs decided jointly.
Sprint Planning + Review ceremonies + bi-weekly iteration demos	Sprint Planning (Wk 1, 5, 9) + Review (Wk 4, 8, 12) + iteration demos (Wk 2, 6, 10). HMGCC SME input minimum at Planning and Review.	HMGCC security researchers run the latest build against real tear-down tasks at every review; findings feed the next sprint backlog.
Mid-project on-site review at HMGCC Milton Keynes	Wk 6, 1 day	Deep-dive on verifier output, bias flags, multilingual coverage; re-prioritise Sprint 2 / Sprint 3 backlog if needed.
Joint go/no-go gates + risk & scope board	End of Sprint 1 (Wk 4) and Sprint 2 (Wk 8); shared register continuous	Formal review against Section 3 acceptance criteria. If a minimum metric is missed by > 20 %, scope is renegotiated jointly, not silently descope. Stretch metrics tracked but not gate-blocking.

Where we would value HMGCC's input: (1) researcher workflow & pain points beyond Alicia; (2) trusted-source register ranking; (3) data-handling constraints (OFFICIAL, BPSS); (4) priority ICS product families & languages.

5. Timescale: 12-week agile sprint plan (3 x 4-week sprints)

Following HMGCC's stated 3 x 4-week cadence: formal review demos at Wk 4, 8, 12 plus bi-weekly iteration demos at Wk 2, 6, 10.

Sprint & weeks	Outcomes	Demo / co-creation moment
Sprint 1: Foundation + Multi-modal & multilingual ingest Wk 1 to 4 (with bi-weekly iteration demo at Wk 2)	Kick-off on-site at HMGCC Milton Keynes; baseline workbench installed in target environment; threat model + architecture sign-off; benchmark harness defined; OCR + handwriting + schematic extraction; full HMGCC format coverage (Word / Excel / PowerPoint / draw.io); offline language detection + translation models packaged for HMGCC priority languages (DE, JA, FR, ZH).	On-site kick-off (signed Phase 1 plan); Sprint 1 review demo at end of Wk 4 with HMGCC researcher pool : schematic + non-EN datasheet, Inventory populated.
Sprint 2: Verifier + Adaptive UX & cross-lingual retrieval Wk 5 to 8 (with bi-weekly iteration demo at Wk 6)	Agentic verification + contradiction detection + on-wire cross-check; calibrated confidence; bilingual ICS-term dictionary curated with HMGCC; long-term memory schema; researcher-profile + proactive surfacing; cross-lingual retrieval (EN query, non-EN source); persistent multi-week threads validated.	Mid-project on-site review at HMGCC Milton Keynes (Wk 6) ; hallucination demo; Sprint 2 review demo at end of Wk 8: 3-week 'Alicia' walkthrough with HMGCC researcher pool.
Sprint 3: Hardening, bias layer + Handover Wk 9 to 12 (with bi-weekly iteration demo at Wk 10)	Bias-flag layer; independent security review; Tier 1 edge-laptop performance tuning; TRL 6 in-environment test plan executed; final benchmarks; documentation, operator guide, IP register, signed CycloneDX SBOM, code handover; Phase 2 roadmap with costings.	TRL 6 test report + signed binaries (Wk 10); Final on-site demonstration session at HMGCC Milton Keynes (Wk 12) ; code & documentation handover to HMGCC engineering team.

6. Budget: £60,000 (all-inclusive ceiling), justified

£60,000 all-in ceiling: time, materials, hardware, overheads, indirects. **Safetech Global Limited is not UK VAT registered, so no VAT applies.** All Phase 1 software components are open-source (Apache 2.0 / MIT); no commercial licence fees. Daily rates blended UK-market for security R&D specialists.

Cost line	Day rate (£)	Days	Subtotal (£)	Sprints	% Total
Technical Lead / Architect & Project Delivery (Fatma, founder rate, full-time)	240	60	14,400	S1 to S3	24.0 %
Senior Cybersecurity & AI Verification Engineer (Jon, 0.5 FTE across all 12 weeks, verifier focus in Sprint 2)	600	30	18,000	S1 to S3	30.0 %
Cybersecurity Advisor (Alan, 1 day / week)	800	12	9,600	S1 to S3	16.0 %
Labour subtotal		102	42,000		70.0 %
OTShield dev / eval workstation (1x RTX 5090 32 GB GDDR7) for Tier 3 internal benchmarking, model selection & calibration; not part of HMGCC deliverable	n/a	n/a	5,500	S1 to S3	9.2 %
Air-gapped test laptop (Windows 11 + Ubuntu latest dual-boot)	n/a	n/a	2,000	S1 to S3	3.3 %
Local LLM tooling, Gemma 4 deployment, signed-installer toolchain	n/a	n/a	900	S1 to S3	1.5 %
Independent security review (external)	n/a	n/a	2,500	S3	4.2 %
Project management & overheads (delivery role combined with Tech Lead)	n/a	n/a	800	S1 to S3	1.3 %
Travel & on-site engagement at HMGCC Milton Keynes (kick-off + mid-project + final demo / handover)	n/a	n/a	1,300	S1 to S3	2.2 %
Reproducible test corpus + benchmark infrastructure (TRL 6 evidence pack)	n/a	n/a	5,000	S1 to S3	8.3 %
TOTAL (all-inclusive)			£60,000		100 %

Value-for-money rationale

- **Founder full-time at £240/day** (vs. £750+ commercial rate) = ~£30,000 real cash discount; **pre-existing OTShield IP at no charge** (Workbench, deep-extract pipeline, offline LLM stack, PCAP / decoy / asset platform, live threat-intel feed). HMGCC pays only for the new capability on top.
- **Chartered specialist + senior assurance.** Jon Medvenics (ChCSP, ex-MoD / NHS Digital / PDS) leads verifier & AI red-team; Alan Jenkins (ex-RAF Squadron Leader, former Babcock CISO) provides 1 day / week of architecture and threat-model assurance.
- **Right-sized hardware + realistic 70 / 30 labour-non-labour split.** The RTX 5090 funds Tier 3 internal benchmarking and calibration only (not the deliverable). The Phase 1 deployment target is a 4 to 8 GB VRAM laptop (Tier 1) plus optional escalation to HMGCC's hosted OpenAI-style endpoint (Tier 2); no cloud dependency, no closed-source model lock-in. Non-labour also funds external security review, on-site engagement and TRL 6 evidence pack.

7. Team

OTShield is a research-first ICS security company. ICS / OT subject-matter authority for the Alicia use case is shared across the named team, not delegated to a separate role: Fatma (8+ years OT / ICS protocols, NATO Locked Shields 2019), Jon (SCADA engineering audits, classified-network exposure), and Alan (former Babcock CISO, UK Defence Intelligence School). *Full CVs and organisational profile in Annex C, outside the 6-page limit.*

Role	Person	Relevant expertise
Technical Lead & Project Delivery	Fatma Erturk Project Lead; Founder & CEO, OTShield	Founder & lead architect of OTShield, accountable for Phase 1 delivery and HMGCC interface. 14+ years software engineering (SCADA, smart-city security, Java microservices); Vodafone Scrum Master Award (2023-24) ; 8+ years OT / ICS protocol expertise (Modbus, IEC 104, DNP3, S7); SCADA Blue Team at NATO Locked Shields 2019 ; Best SCADA / ICS / OT Security Training Provider 2024; Turkish Presidency of Defence Industries Woman Entrepreneur Award.
Senior Cybersecurity & AI Verification Engineer	Jon Medvenics ChCSP, OSCP, GXPIN, GREM	UK Chartered Cyber Security Professional, founder of Heretek Ltd. Government delivery: Ministry of Defence (classified-network SOC), Ministry of Justice (Red Team Operator), NHS Digital (COVID-19 incident response), Parliamentary Digital Service . Specialist in AI / LLM red-teaming and verification with Microsoft's PyRIT.
Cybersecurity Advisor	Alan Jenkins CISM, PRINCE2, CISP	35+ years cyber / enterprise security; 20+ years senior RAF Police officer (Squadron Leader); graduate of UK Defence Intelligence & Security School. Former CISO at Babcock International (UK defence prime), T-Systems UK, CSC UK; former IBM Security Associate Partner; long-standing CISO-in-Residence at CyLon. Independent assurance over architecture, threat model, data-handling and HMGCC engagement.

8. Exploitation, Phase 2 & risk

Dual-use exploitation: HMGCC is one customer in a wider market

OTShield is a commercial dual-use product. We are pre-revenue but commercially defined: active pipeline, three published price tiers, an internet-exposed decoy fabric validating the platform against real attackers today. The Phase 1 capability is directly reusable across the segments below, so HMGCC inherits a tool sustained by Safetech Global's wider commercial activity.

- **UK national security & defence (beyond HMGCC):** other government departments running ICS tear-downs and supply-chain assurance, including MOD, NCSC, NPSA and defence primes.
- **Critical infrastructure operators (largest commercial segment):** energy, water, transport, pharma. Three price tiers: Starter (£20k/yr, 50 assets), Growth (£50k/yr, 500 assets), Advanced (£75k/yr, unlimited). NIS2 / IEC 62443 / NCSC CAF aligned; coexists with Dragos / Claroty. *Full partner-channel evidence in Annex E.*
- **ICS vendor red teams & international allies:** Siemens-, Rockwell-, Schneider-class internal tear-down programmes; Five Eyes & NATO partner governments. Already selected for **Portuguese Air Force InnCyber** and **Hubgrade Cyber (Germany)**, plus UK accelerators (CyLon, Digital Catapult Cyber 101, Cyber Runway).
- **Phase 2 vision, subject entirely to HMGCC's discretion based on Phase 1 outcomes (no commitment for Phase 2):** if HMGCC chooses to proceed, an indicative scope of approx. £250 to £400 k over 6 to 9 months would deliver (i) signed offline-update pipeline productionised, fed by the live decoy-fabric threat-intel feed; (ii) multi-researcher collaborative workspaces with shared Findings ledger; (iii) hardware-in-the-loop firmware extraction (JTAG / SWD); (iv) REST / gRPC API + SIEM-style export for HMGCC analyst infrastructure. **The Phase 1 deliverable is fully self-sufficient under the royalty-free licence; Phase 2 is not a precondition for HMGCC to retain and use the Phase 1 output.**

Risk register (top 6)

Risk	Lhood	Impact	Mitigation
Tier 3 dev-workstation procurement slip (RTX 5090)	Low	Low	RTX 5090 funds internal benchmarking and Tier 3 calibration <i>only</i> ; the HMGCC deliverable is the Tier 1 edge laptop binary plus the optional Tier 2 HMGCC-endpoint integration, neither of which depend on the dev workstation. Used RTX 5090 / RTX 4090 24 GB are equivalent fallbacks; cloud-GPU dev environment available as a temporary backstop. Project schedule is unaffected by procurement variance.
Local LLM accuracy on technical ICS jargon insufficient	Med	High	Domain-tuned prompts + verifier agent + protocol-aware extractors already operational; benchmark-driven model selection in Sprint 1; AI red-team validation by Jon Medvenics using PyRIT-class methodology.
Hallucination / over-confident answers	Med	High	Mandatory verifier pass + on-wire cross-check + calibrated confidence + 'needs more sources' flag; pre-publication gate.
Tier 1 edge-laptop performance (4 to 8 GB VRAM)	Med	Med	Tier 1 primary models (Gemma 4 9B Q4_K_M for 4 GB, 13B Q4_K_M for 8 GB) sized explicitly for HMGCC's 4 to 8 GB VRAM envisaged laptop. Llama 3.2 1B retained as CPU-only fallback for VRAM-constrained or CPU-only edge cases. Tier 2 HMGCC endpoint absorbs higher-accuracy verifier passes when edge confidence falls below threshold. Latency target ≤ 30 s Tier 1 / ≤ 8 s Tier 2 (see Annex A.8).
HMGCC test data delivery slip	Low	Med	HMGCC will deliver representative test documents incrementally during the first 4 weeks. OTShield's internal ICS test corpus is used as fallback in Sprint 1 to keep the schedule on track until HMGCC data arrives.
Phase 2 funding not secured before Phase 1 ends	Med	Low	Safetech Global's commercial roadmap and active customer pipeline sustain the codebase even if Phase 2 grant slips; Phase 2 features are then prioritised by the order in which the pipeline converts. The Phase 1 deliverable remains usable to HMGCC under the royalty-free licence whether or not Phase 2 proceeds.

Live demo (recorded walkthrough)

A recorded end-to-end walkthrough of the OTShield platform is available on YouTube and shows everything the main proposal claims as *operational today*: SOC Mode and the live decoy fabric attacker map with the 157-attacker / 20-country / 12,968-attack figures, and Research Mode (Bundles, Library, Summary, Inventory, Threads with citations and HIGH / MEDIUM / LOW confidence labels). Reviewers can verify every "today" claim in this proposal against the demo before reading the annexes.

Demo video: <https://www.youtube.com/watch?v=2QdR3wRGQhk>

We are not building a better search tool. We are delivering a TRL 6 cognitive layer for ICS vulnerability research, purpose-built, offline-native, evidence-driven. The platform exists today; the £60k unlocks the verification, adaptive and multi-modal capabilities that turn it into the personal assistant Alicia needs.